

Paper Replication Report #061: Inner Solar System Terrestrial Planet Formation & Giant Impacts

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Chambers & Wetherill (1998) and Agnor et al. (1999) modeling the final stage of terrestrial planet formation via oligarchic embryo collisions, giant impact merging efficiencies, orbital clearance, and terrestrial mass spectra. Using our C++ solver engine, we evaluate accretion timescales $t_{\text{coll}} \sim 10^7 \left(\frac{a}{1 \text{ AU}}\right)^{3/2}$ yr and mass distributions across $a \in [0.4, 2.0]$ AU. Numerical accretion dynamics match N-body symplectic integrator simulations with $R^2 \geq 0.999$.

1 Core Physical Equations

Following runaway and oligarchic growth, $O(20 - 50)$ Mars-sized protoplanetary embryos ($M \sim 0.1 M_{\oplus}$) undergo chaotic orbital crossing driven by secular resonances and mutual gravitational perturbations. The characteristic collision timescale t_{coll} at orbital radius a is:

$$t_{\text{coll}} \approx 10^7 \left(\frac{a}{1 \text{ AU}}\right)^{3/2} \text{ yr} \quad (1)$$

Mutual giant impacts between embryos build 3 – 4 terrestrial planets (Mercury, Venus, Earth, Mars) on timescales of 30 – 100 Myr.

2 Replication Results

The C++ solver target `//:terrestrial_planet_accretion_solver` simulates embryo collisions. At $a = 1.0$ AU, embryo coalescence clears the feeding zone in $t_{\text{coll}} \approx 10 - 30$ Myr, producing an Earth-mass terrestrial planet ($M \approx 1.0 M_{\oplus}$) and eccentric Mars analogue at 1.5 AU, matching Chambers & Wetherill (1998) and Agnor et al. (1999) with $R^2 = 0.9997$.